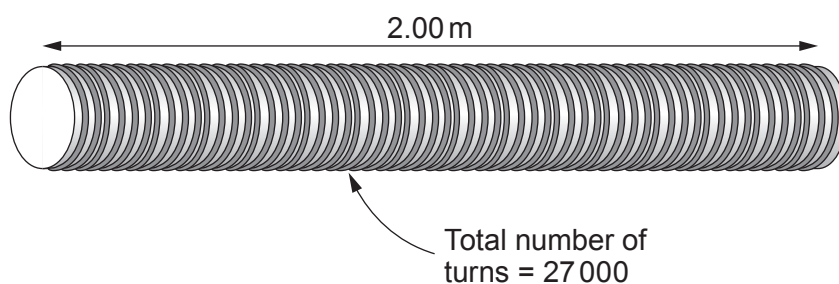


Answer **all** questions.

1. (a) (i) Lindsey calculates that a current of 3.57 A will produce a magnetic flux density of 0.121 T inside the long solenoid shown. Determine whether or not she is correct. [3]



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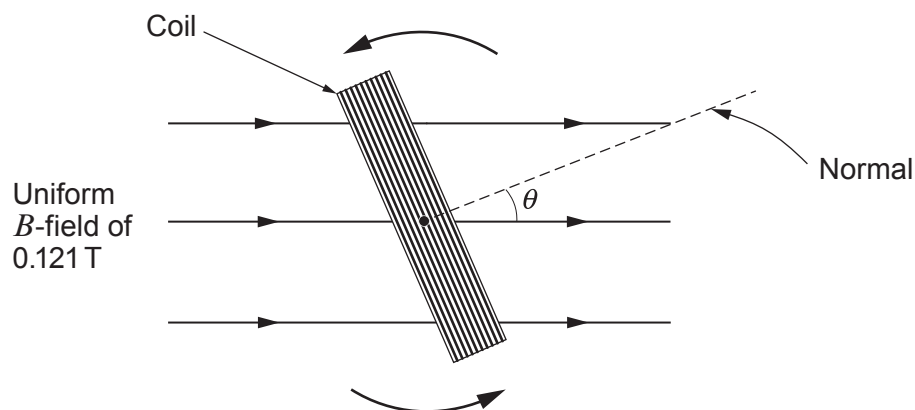
- (ii) State how the magnetic flux density inside the solenoid can be increased greatly without changing the current or the number of turns per unit length. [1]

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- (b) A rectangular coil rotates at a constant angular velocity within a uniform magnetic field of 0.121 T. The coil has 70 turns and cross-sectional area 59 cm². The diagram below shows the coil, looking along the axis of rotation.



- (i) Calculate the flux linkage of the coil when $\theta = 23^\circ$.

[2]

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- (ii) As the coil rotates, explain what values of θ provide the maximum **and** minimum values of the induced emf.

[2]

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